



MIT Art, Design and Technology University, Pune
MIT School of Computing,
Department of Information Technology

S.Y. B. Tech. Semester-II

Laboratory Manual

NAME OF COURSE: WEB PROGRAMMING
YEAR & SEM: S.Y- B.TECH. SEMESTER-II
AY.2025 – 2026

Student Name	Dipak Nagnath Munde
Enrollment No.	ADT24SOCB1711
Class / Division	SY IT (DA-1)
Project Title	Sports equipment website
Course Coordinator	Prof. Rohini Bhosale
Practical Coordinator	Prof. Rohini Bhosale

Web Programming SEMESTER – IV			
Course Code:	23IT2006	Course Credits:	02
Teaching Hours / Week (L:T:P):	0:0:4	CA Marks:	25
Total Number of Teaching Hours:		END-SEM Marks:	25
Course Pre-requisites:			
Course Description: This course provides a comprehensive introduction to web technology, designed to help students develop a strong foundation in building and managing websites and web applications. The curriculum covers key topics such as HTML, CSS, and JavaScript, PHP, MySQL, which are essential for creating interactive, well-designed web pages. Students will also explore the principles of responsive design, ensuring that web applications are optimized for different devices and screen sizes. The course dives deeper into server-side technologies, including HTTP, web servers, and databases, allowing students to understand how websites function behind the scenes. Emphasis is placed on practical learning, and students will gain hands-on experience by working on projects that showcase their ability to design, develop, and deploy websites. By the end of the course, students will be proficient in using modern web technologies to create web applications. They will understand how to handle client-server interactions, manage user data, and implement various web technologies to enhance the functionality of their applications.			
Course Learning Objectives: This course will enable the students to: 1. Understand fundamental concepts of front-end web development. 2. Enable students to create basic web pages incorporating essential elements such as images, hyperlinks, lists, tables, and forms. 3. Teach students how to use CSS to manage fonts, lists, colors, text alignment, and background images for a cohesive and aesthetically pleasing web design. 4. Develop an understanding of JavaScript scopes to manage the visibility and lifetime of variables and functions effectively. 5. Equip students with the skills to implement and handle JavaScript events, enabling enhanced user interactions through event-driven programming. 6. Apply comprehensive knowledge of HTML, CSS, and JavaScript to develop a complete front-end application. Utilize project-based learning to showcase problem-solving skills and creativity in web development projects. 7. Configure server environments with Apache/TOMCAT. 8. Set up a PHP development environment and write basic PHP scripts. 9. Master PHP programming constructs for web development tasks. 10. Create and process HTML forms, and manage MySQL database operations. 11. Develop comprehensive back-end applications using PHP and MySQL.			
Course Outcome: After taking this course, Students will be able to : 1. Apply knowledge of HTML to create the structure of the webpage and CSS to style and layout the elements, making the application visually appealing. 2. Apply comprehensive knowledge of HTML, CSS, and JavaScript to develop a complete front-end application and utilize project-based learning to showcase problem-solving skills and creativity in web development projects. 3. Set up and configure a server environment using tools like Apache or TOMCAT and set up a PHP development environment. Write & execute simple PHP scripts, understanding PHP syntax and basic features, create HTML forms to collect user data and integrate with PHP for			

processing.

4. Design and develop a back-end application using PHP and MySQL, implementing CRUD operations to manage data effectively.

UNIT - I	Introduction to HTML and Cascading Style Sheet	09 Hours
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Module 1 - Markup Language (HTML): Introduction to HTML, Formatting and Fonts, Commenting Code, Anchors, Backgrounds, Images, Hyperlinks, Lists, Tables, Frames, HTML Forms

Module 2 - CSS: Need for CSS, introduction to CSS, basic syntax and structure, Levels of style sheets, Style specification formats, BOX Model, Selector forms, Property value forms, Font properties, List properties, Color, Alignment of text, Background images

Pedagogy	ICT Teaching / PowerPoint Presentation and Videos: Use tools like Visual Studio Code (free). Videos: https://www.coursera.org/learn/html-css-javascript-for-web-developers
	Self-study / Do it yourself /: Practice creating basic HTML pages and enhancing them using CSS.
	Experiential Learning Topics: Design a simple webpage for coffee shop website
	PBL - Project Based Learning: Create a multi-page website (e.g., coffee shop website) using HTML and CSS.

UNIT - II	Front-End Development	09 Hours
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Module 3 - Overview of JavaScript, including JS in an HTML (Embedded, External), Basic JS syntax, basic interaction with HTML

Module 4 - Core features of JavaScript: Data types, Control Structures, Arrays, Functions and Scopes

Pedagogy	ICT Teaching / PowerPoint Presentation and Videos: Use tools like Visual Studio Code (free). Videos: https://www.coursera.org/learn/javascript-basics
	Self-study / Do it yourself /: Solve exercises on JavaScript syntax, control structures, and functions
	Experiential Learning Topics: Build a web page with interactive elements (e.g., a simple calculator).
	PBL - Project Based Learning: Develop an interactive webpage that uses JavaScript to validate form inputs or perform basic calculations.

UNIT - III	Advanced Front-End Development	09 Hours
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Module 5 - DOM: DOM levels, DOM Objects and their properties and methods, Manipulating DOM

Module 6 - JavaScript Events: JavaScript Events, Types of JavaScript Events, Objects in JS, Event Handling

Pedagogy	ICT Teaching / PowerPoint Presentation and Videos: https://www.coursera.org/learn/building-interactive-web-pages-using-
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	javascript Use tools like Visual Studio Code (free).
	Self-study / Do it yourself /: Practice exercises on DOM traversal and event handling.
	Experiential Learning Topics: Add dynamic behavior to a webpage using DOM and events (e.g., a to-do list app).
	PBL - Project Based Learning: Develop a web page with dynamic content (e.g., a task manager or interactive quiz) using DOM manipulation and event handling.

UNIT - IV	Server Side Scripting	09 Hours
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Module 7 - Set up and configure a server environment using tools like Apache or TOMCAT, set up a PHP development environment.

Module 8 - Introduction to PHP: : Introduction to PHP, Server side scripting Vs Client side scripting, Basic Development Concepts (Mixing PHP with HTML), Creating, Writing & Running First PHP Script, PHP syntax, conditions & Loops, Functions, String manipulation, Arrays & Functions,

Module 9 - Form handling with HTML and PHP: Designing of Forms using HTML, Form Handling using GET and POST methods of Form

Pedagogy	ICT Teaching / PowerPoint Presentation and Videos: https://www.coursera.org/learn/web-applications-php Use tools like Visual Studio Code (free), XAMPP/WAMP for PHP server setup, and MySQL Workbench for database management
	Self-study / Do it yourself /: Practice exercises on form handling and server-side scripting with PHP.
	Experiential Learning Topics: Create a basic form for data submission and handle it using PHP (e.g., feedback form).
	PBL - Project Based Learning: Develop a small server-side application (e.g., a contact form with email validation and submission).

UNIT - V	Working with Databases and Web Application Development	09 Hours
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Module 10 - Working with databases using MySQL with PHP: MySQL database, create database, create table, primary key with AUTO_INCREMENT setting, Insert Data Into a Database Table, Select Data From a Database Table, Open or close a Connection to the MySQL Server.

Module 11 - Web Application Development (Project): Develop the web application to handle client-server interactions, manage user data, and implement various web technologies to enhance the functionality of their applications. Example: Website for a Coffee Shop

Pedagogy	ICT Teaching / PowerPoint Presentation and Videos: Use tools like Visual Studio Code (free), XAMPP/WAMP for PHP server setup, and MySQL Workbench for database management Videos: https://www.coursera.org/learn/web-app
	Self-study / Do it yourself /:

	Exercises on creating and manipulating databases using PHP and MySQL.
	Experiential Learning Topics: Create a database and design a webpage to display its data dynamically.
	PBL - Project Based Learning: Develop a fully functional web application (e.g., a Coffee Shop website or e-commerce platform) that integrates database functionality for data management.

Text Books:

1. "HTML and CSS: Design and Build Websites" by Jon Duckett.
2. "Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics" by Jennifer Niederst Robbins.
3. Achyut Godbole & Atul Kahate, ||Web Technologies: TCP/IP to Internet Application Architectures||, McGraw Hill Education publications, ISBN, 007047298X, 9780070472983.
4. Ralph Moseley & M. T. Savaliya, —Developing Web Applications||, Wiley publications, ISBN 13 : 9788126538676.

Reference Books:

1. Eloquent JavaScript: A Modern Introduction to Programming by Marijn Haverbeke.
2. JavaScript: The Good Parts by Douglas Crockford.
3. CSS Secrets: Better Solutions to Everyday Web Design Problems by Lea Ver.
4. Web Technologies- Jeffery C. Jackson, ISBN 978-81-317-1715-8 Pearson 2015.
5. PHP Objects, Patterns, and Practice by Matt Zandstra
6. MySQL Cookbook by Paul DuBois.
7. Advanced PHP Programming - George Schlossnagle- ISBN 0-672-32561-6,2004.

URLs (Optional) - List of Online Courses

1. W3Schools HTML, CSS, JavaScript Tutorial: <https://www.w3schools.com/html/>
2. Mozilla Developer Network (MDN) Web Docs - HTML, CSS, JavaScript, DOM: https://developer.mozilla.org/en-US/docs/Learn/HTML/Introduction_to_HTML
3. Project-Based Learning Resources: <https://developer.mozilla.org/en-US/docs/Learn>

Contents beyond Syllabus:

1. Web Essentials
2. Using JavaScript to handle form submission and login events (e.g., onsubmit, onclick)
3. JavaScript Form validations, General Input Validation, Password Validation
4. Storing user data (like a username) temporarily using localStorage or sessionStorage
5. Dynamically updating the content of the webpage, such as displaying a welcome message
6. Redirecting users using window.location

List of Experiments:

In this series of assignments, you will create a pet shop / any other website step by step. Each assignment will focus on a different aspect of the website, covering various HTML elements, CSS, JavaScript, PHP and MySQL concepts.

Laboratory/Project Assignment Guidelines:

1. Project Selection:

- Each student must select a unique project topic for their laboratory assignments.
- The chosen project topic should align with the concepts covered in the course syllabus. The chosen project topic should be approved by the course coordinator/ subject teacher.
- Students have the freedom to choose their project topics based on their interests and career aspirations.
- Project topics may include but are not limited to:
 - E-commerce website
 - Blogging platform
 - Online booking system
 - Content management system (CMS)
 - Discussion forum
 - Social networking platform
 - Task management application
 - Portfolio website

2. Laboratory Assignments:

- Throughout the course, students will complete laboratory assignments related to their chosen project topic.

3. Evaluation Criteria:

- The laboratory assignments and the final project will be evaluated based on criteria such as Structure and Semantics, Content Organization, Forms and Inputs, Links and Navigation, Styling and Layout, Design Consistency, Functionality, Code Quality and adherence to project requirements.
- Students are expected to demonstrate creativity, and a comprehensive understanding of web development principles in their projects.
- The laboratory assignments based on chosen project topics will be assessed based on several key criteria that reflect both technical proficiency and creative application in web development. These include:
 - Structure & Semantics: Proper use of HTML to create a logical, accessible structure with meaningful and semantically correct elements.
 - Content Organization: Clear and intuitive organization of content, ensuring ease of navigation and logical flow throughout the site.
 - Forms & User Input: Effective implementation of forms and user input elements that are functional, validated, and accessible.
 - Links & Navigation: Well-structured navigation and functional links that provide a seamless user experience.

- Styling & Layout: Visually appealing and responsive design, with a well-executed layout that adapts to various screen sizes.
- Design Consistency: Uniformity in design elements, including colors, typography, and spacing, to maintain a cohesive look and feel across the site.
- Functionality: Full functionality of all interactive elements, ensuring a bug-free, smooth experience for users.
- Code Quality & Best Practices: Clean, well-organized, and efficient code that adheres to modern web development best practices and is easy to maintain.

4. Submission and Presentation:

- The project and project report/journal must be submitted within the specified deadline and should meet the specified requirements outlined by the course coordinator/ subject teacher.

Project Problem Statement-

Design and develop a basic website for a local pet shop using HTML, CSS, JavaScript, PHP and MySQL. This website will serve as an online presence for the pet shop, effectively communicating the brand identity, showcasing the menu, providing essential information, and allowing customers to easily get in touch or locate the shop.

The project directory is as follows:

pet-shop-website/

```
├── css/
│   └── styles.css
├── html/
│   ├── index.html
│   ├── menu.html
│   ├── about.html
│   └── contact.html
├── js/
│   └── scripts.js
└── images/
```

1.	Create the basic structure of the coffee shop website, including the home page layout with a header, main content area, and footer. Prepare a common project website design and plan document for all assignments. Consider following points: 1. Brief information about the project. 2. Set the goals & deliverables. 3. Finalize the modules of the project. 4. Define the audience.
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	5. Describe pain points & the ideal experience (On the basis of existing systems) 6. Set the visual direction 7. Map out the Project structure. 8. Plan the content for each page. 9. Add ideas for content, images & layout. 10. Determine your site structure or Create content for your core website pages: a. Home page b. About page c. Product/Service page d. Testimonial/review page e. Contact page f. Starter blog posts 11. Create and collect design elements These design elements define your brand personality and help customers feel what your brand represents through the use of: a. Colors b. Fonts and typography c. Logos d. Images and photos
2.	HTML A. Create a detailed home page for the coffee shop website. B. Create a detailed menu/product page for the coffee shop website, listing all available items categorized appropriately. C. Create a cart page that allows customers to review and manage the items they wish to purchase before proceeding to checkout. D. Create an about us page that provides detailed information about the coffee shop's history, mission, and team. E. Create a contact page that allows customers to easily get in touch with the coffee shop through a form. F. Design and implement admin/user registration form for the coffee shop website. G. Design and implement admin/user login form for the coffee shop website.
3.	CSS A. Enhance the layout of the coffee shop website using CSS Grid for the home page. B. Use CSS Grid to layout the menu/product items in a structured and style the

	menu categories with appropriate headings, spacing, separators, images, descriptions, and prices.
4.	CSS A. Enhance the cart page to make it user-friendly and visually appealing. Style the cart items with appropriate margins, paddings, and input field styles to provide a seamless shopping experience. B. Enhance and style the about us page with appropriate margins, paddings, and input field styles. C. Enhance and style the contact page to make it user-friendly and visually appealing. Style the contact form with appropriate margins, paddings, and input field styles. D. Enhance and style the admin/user registration form with appropriate margins, paddings, and input field styles. E. Enhance and style the admin/user login form with appropriate margins, paddings, and input field styles.
5.	JavaScript A. Implement user registration and login forms for the coffee shop website. These forms will allow users to create an account, log in, and access personalized features, such as saving favorite items or viewing order history. User Registration Form will allow new customers to sign up and create an account on the website. The form will capture basic user details, including the name, email address, and password (not limited to these fields). User Login Form will allow registered users to log into their accounts. The form will require an email address and a password to authenticate the user. B. Provide validations for user registration and login forms to validate the input to ensure that all required fields are filled and that the email format is valid. (Contents beyond Syllabus) C. Develop cart functionality to allow users to add items, update quantities, and remove items.
6.	JavaScript A. The user login form will allow registered users to log into their accounts. The form will require an email address and a password to authenticate the user. B. If the login is successful, the user should be redirected to the homepage or their user dashboard. (Contents beyond Syllabus) C. Use localStorage or sessionStorage to store authentication data, such as the user's email and login status. This ensures that once a user is logged in, they remain authenticated even after the page reloads or when they visit the site again. (Contents beyond Syllabus) D. Save the cart data to local storage when items are added, updated, or removed. Retrieve and load the cart data from local storage when the page loads. (Contents beyond Syllabus)

7.	PHP A. Develop a PHP script to handle user registration for the Coffee Shop website. The script should accept input from users for their name, email address, password, etc. (all required fields for registration). B. Implement error handling to notify users of any issues during registration, such as validation errors. C. Provide feedback to the user upon successful registration, either through a confirmation message or a redirect to a login page.
8.	PHP A. Develop a PHP script to handle user login for the Coffee Shop website. The script should accept input from users for their login credentials. (all required fields for login). B. Provide feedback to the user upon successful login, either through a confirmation message or a redirect to a welcome page. C. Implement error handling to notify users of login failures due to incorrect credentials or other errors. D. Provide feedback to the user upon successful login, either through a welcome user name message or a redirect to a home page.
9.	PHP and MySQL A. Develop a PHP script that allows users to manage their shopping cart for an e-commerce website (e.g., a Coffee Shop store). The script should allow users to add items to their cart, view their cart contents, and remove items if needed. B. Develop a PHP script to manage the shopping cart for an e-commerce website (e.g., a Coffee Shop store) using MySQL. This script should allow users to add items to their cart, view their cart contents, and remove items from the cart. The cart data should be stored in the MySQL database to allow persistence across sessions.
10.	PHP and MySQL A. Develop a PHP script to handle the checkout process for users who are ready to complete their purchase. The script should process the cart data and provide feedback to the user upon successful or failed checkout. B. Develop a PHP script that processes the checkout process for users who are ready to complete their purchase, integrating the MySQL database for handling user and order information. The script should validate user input, process the cart data, and provide feedback upon successful or failed checkout.

Experiment No.1

Problem Statement:

Create the basic structure of the Pet shop website, including the home page layout with a header, main content area, and footer.

Prepare a common project website design and plan document for all assignments. Consider following points:

1. Brief information about the project.
2. Set the goals & deliverables.
3. Finalize the modules of the project.
4. Define the audience.
5. Describe pain points & the ideal experience (On the basis of existing systems)
6. Set the visual direction
7. Map out the Project structure.
8. Plan the content for each page.
9. Add ideas for content, images & layout.
10. Determine your site structure or Create content for your core website pages:
 - a. Home page
 - b. About page
 - c. Product/Service page
 - d. Testimonial/review page
 - e. Contact page
 - f. Starter blog posts
11. Create and collect design elements
12. These design elements define your brand personality and help customers feel what your brand represents through the use of:
 - a. Colors
 - b. Fonts and typography
 - c. Logos
 - d. Images and photos

Objective:

To design the basic structure of a Pet shop website by planning its layout, content, and visual elements, ensuring it meets user needs and effectively represents the brand.

Theory:

Project Design and Plan Document for pet Shop Website

1. Brief Information about the Project

This project involves designing a user-friendly and visually appealing **Pet Shop Website**. The goal is to attract pet lovers and provide them with information about pets, pet products, and services. The website will include features such as product browsing, customer testimonials, contact forms, and

user authentication (login/registration). It aims to enhance user experience and promote the pet shop's brand online.

2. Goals & Deliverables

Goals

- Develop an attractive and responsive pet shop website
- Showcase pets, pet food, and accessories
- Provide smooth navigation and user experience
- Enable login and registration for users
- Build trust through testimonials and contact accessibility

Deliverables

- Home Page
- About Page
- Products/Services Page
- Testimonials Page
- Contact Page
- Login Page
- Registration Page
- Optional Blog Section

1. Modules of the Project

1. Home Page

- Welcome section with banner
- Featured pets/products
- Navigation bar
- Footer

2. About Page

- Shop history
- Mission & vision
- Team introduction

3. Products/Services Page

- Pet categories (Dogs, Cats, Birds, Fish)
- Pet food & accessories
- Product details with images & price

4. Testimonials Page

- Customer reviews
- Ratings
- Review submission form

5. Contact Page

- Contact form
- Address & phone number
- Map location

6. Login Module

- Email & password login
- Forgot password option

7. Registration Module

- User signup form
- Validation

8. Footer Module

- Quick links
- Social media
- Contact details

4. Target Audience

- **Pet Owners** – looking for food, accessories, and services
- **Animal Lovers** – interested in adopting pets
- **Families** – searching for pets
- **Students** – looking for affordable pet options
- **Online Buyers** – prefer ordering pet products online

5. Pain Points & Ideal Experience

Pain Points

- Confusing navigation
- Lack of product details
- No mobile responsiveness
- Poor design
- No online ordering

Ideal Experience

- Simple and clean interface
- Easy navigation
- Mobile-friendly design
- Detailed product information
- Fast and smooth interaction

6. Visual Direction

Color Palette

- Brown (#6F4E37) - warmth
- Cream (#F5F5DC) - background
- Green (#556B2F) - CTA buttons
- White (FFFFFF) - clarity

Typography

- Headings: Poppins / Roboto
- Body: Open Sans

Images

- Pets (dogs, cats, birds)
- Pet food & accessories
- Happy customers with pets

7. Map out the Project structure pet_shop_website/

```
├── index.html      # Home page
├── about.html      # About page
├── products.html   # Products/Services page
├── testimonials.html # Testimonials page
├── contact.html    # Contact page
├── login.html      # Login page
├── register.html   # Registration page
├── blog.html       # Optional blog page
├──
├── assets/
│   ├── css/
│   │   ├── style.css      # Global CSS
│   │   └── responsive.css # Media queries for mobile optimization
│   ├──
│   ├── js/
│   │   ├── main.js        # Interactive scripts
│   │   └── formValidation.js # Scripts for login and registration validation
│   ├──
│   └── images/
│       ├── logo.png      # Website logo
│       ├── homepage_banner.jpg # Hero banner for home
│       ├── pet_menu/      # Images for menu items
│       ├── team_photos/   # Images for the about page
│       └── icons/         # Icons for UI elements
├──
└── fonts/
    ├── Poppins/          # Primary font
    └── OpenSans/         # Secondary font
```

```

└── README.md          # Project documentation file
pet_shop_website/
├── index.html
├── about.html
├── products.html
├── testimonials.html
├── contact.html
├── login.html
├── register.html
├── assets/
├── css/
├── js/
├── images/
└── README.md

```

8. Content Plan for Each Page

Home Page

- Banner: "Find Your Perfect Pet Companion"
- Featured pets/products
- Quick links

About Page

- Story of the pet shop
- Mission & values

Products Page

Pet categories Product listings

Testimonials Page

- Customer reviews
- Ratings

Contact Page

- Contact form
- Address & map

9. Content, Images & Layout Ideas

- Hero image with pets

- Grid layout for products
- Cards for testimonials
- Icons for navigation
- Clean and minimal layout

10. Core Website Pages Content

Home Page

- Welcome message
- Featured pets
- CTA: “Adopt Now”

About Page

- Shop story
- Team

Products Page

- Pets & accessories

Testimonials Page

- Customer feedback

Contact Page

- Inquiry form

Blog Ideas

- Pet care tips
- “How to choose a pet”
- “Best food for dogs & cats”

11. Design Elements

Colors

- Earthy and natural tones

Fonts

- Simple and readable

Logo

- Pet icon (paw or animal shape)

Images

- High-quality pet photos

Conclusion:

The pet Shop Website project serves as a comprehensive exercise in understanding and applying web development fundamentals. Through careful planning and implementation of design elements, content organization, and user experience considerations, the assignment ensures a functional, visually appealing, and user-friendly platform.

By creating essential pages like the Home, About, Product/Service, Testimonial, Contact, and Login/Registration pages, the website provides customers with intuitive navigation and engaging content. The visual direction, including the use of harmonious colors, typography, logos, and images, helps reinforce the brand's identity and appeal to the target audience.

This assignment highlights the importance of a structured approach to web design—starting with setting goals and mapping out the project structure, followed by planning and designing the content. By addressing audience needs and incorporating user-centric solutions, the project enhances the online presence of a pet shop while delivering a seamless digital experience for customers.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<title>Pet Shop - Home</title>
<style>
/* Reset */
* {
margin: 0;
padding: 0;
box-sizing: border-box;}
body {
font-family: Arial, sans-serif;}
/* Header */
header {
background-color: #6F4E37;
color: white;
padding: 15px;
text-align: center;}
/* Navigation */
nav {
background-color: #3C2F2F;
padding: 10px;
text-align: center;}
nav a {
color: white;
margin: 0 15px;
text-decoration: none;
font-weight: bold;}
```

```

nav a:hover {
color: #F5F5DC;}
/* Main Content */
.container {
padding: 20px;
background-color: #F5F5DC; min-
height: 400px;}
.hero {
text-align: center;
padding: 40px;
background-color: #FAF3E0; border-
radius: 10px;}
.hero h1 {
margin-bottom: 10px;}
.products {
margin-top: 30px;
display: flex;
justify-content: space-around;}
.card {
background: white;
padding: 15px;
width: 30%;
text-align: center;
border-radius: 10px;
box-shadow: 0px 0px 5px gray;}
.card img {
width: 100%;
height: 150px;
border-radius: 10px;}
/* Footer */
footer {
background-color: #6F4E37;
color: white;
text-align: center;
padding: 10px;}
</style>
</head>
<body>
<!-- Header -->
<header>
<h1> Happy Paws Pet Shop</h1>
<p>Your One Stop Pet Destination</p>
</header>
<!-- Navigation -->
<nav>
<a href="#">Home</a>
<a href="#">About</a>
<a href="#">Products</a>
<a href="#">Testimonials</a>

```

```

<a href="#">Contact</a>
<a href="#">Login</a>
</nav>
<!-- Main Content -->
<div class="container">
<!-- Hero Section -->
<div class="hero">
<h1>Find Your Perfect Pet Companion</h1>
<p>We provide pets, food, and accessories for your lovely friends.</p>
</div>
<!-- Products Section -->
<div class="products">
<div class="card">

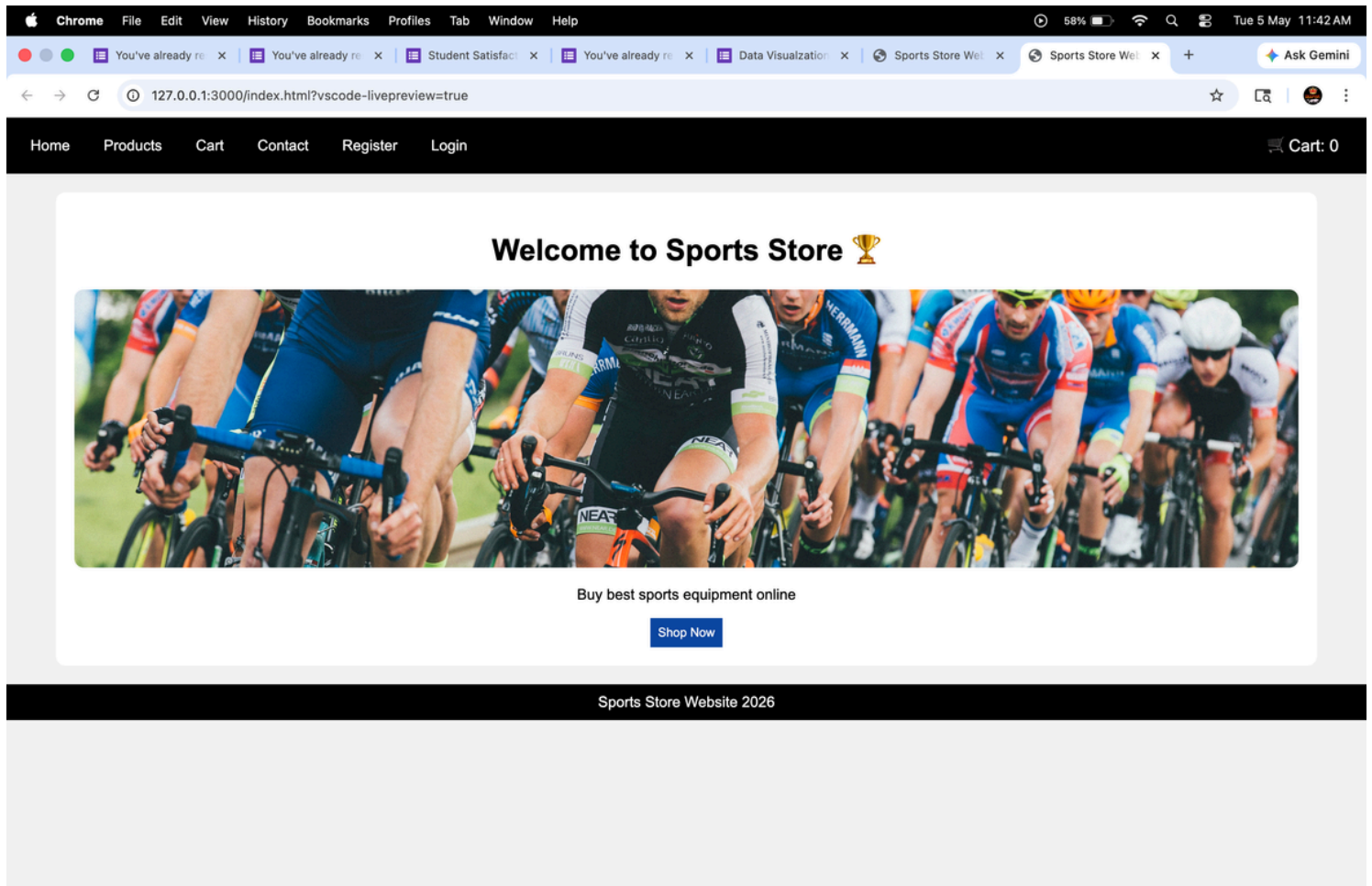
<h3>Dogs</h3>
<p>Friendly and loyal companions.</p>
</div>
<div class="card">

<h3>Cats</h3>
<p>Playful and adorable pets.</p>
</div>
<div class="card">

<h3>Birds</h3>
<p>Colorful and cheerful friends.</p></div></div></div>
<!-- Footer --><footer>
<p>© 2026 Happy Paws Pet Shop | Contact: 9876543210</p>
</footer>
</body>
</html>

```

Output :



Topic: Introduction to HTML and CSS

Experiment No.2,3,4

Problem Statement:

1. HTML

- A. Create a detailed home page for the Pet Shop website.
- B. Create a detailed product/menu page for the Pet Shop website, listing pet products and accessories in categorized form.
- C. Create a cart page that allows customers to review, update, and manage selected products before checkout.
- D. Create an About Us page providing detailed information about the Pet Shop's history, mission, vision, and team members.
- E. Create a Contact page that allows customers to communicate with the Pet Shop using a contact form.
- F. Design and implement Admin/User Registration forms for the Pet Shop website.
- G. Design and implement Admin/User Login forms for the Pet Shop website.

2. CSS

- A. Enhance the layout of the Pet Shop website using CSS Grid for the home page.
- B. Use CSS Grid to arrange product items in a structured layout and style product categories with proper headings, spacing, images, descriptions, and prices.

3. CSS

- A. Enhance the cart page to make it visually appealing and user-friendly. Style the cart items with proper margins, paddings, buttons, and input fields.
- B. Enhance and style the About Us page with suitable layouts, fonts, colors, and spacing.
- C. Enhance and style the Contact page and contact form using proper margins, paddings, and input field styles.
- D. Enhance and style the Admin/User Registration form with attractive layouts and input field designs.
- E. Enhance and style the Admin/User Login form with responsive design and suitable styling.

Objective:

To design and develop a responsive and visually attractive Pet Shop website using HTML and CSS.

Theory:

HTML (HyperText Markup Language) and CSS (Cascading Style Sheets) are the fundamental technologies used for designing and developing web pages. HTML provides the basic structure and content of webpages, while CSS is used for styling, layout management, responsiveness, and visual presentation.

By combining HTML and CSS, interactive and user-friendly websites can be developed efficiently. In this experiment, a Pet Shop website is created containing multiple pages such as Home, Products, Cart, About Us, Contact, Registration, and Login pages.

What is HTML?

HTML is a markup language used to create the structure of web pages. It defines various webpage elements such as headings, paragraphs, images, forms, tables, and hyperlinks.

HTML allows developers to:

- Create structured web documents.
- Add text, images, tables, lists, and multimedia content.
- Link multiple pages together.
- Design forms for user interaction.
- Create navigation systems for websites.

HTML uses tags to define webpage elements.

What is CSS?

CSS (Cascading Style Sheets) is used to style and design HTML webpages. It controls webpage appearance including colors, fonts, layouts, spacing, responsiveness, and animations.

CSS helps to:

- Improve website appearance.
- Create responsive layouts.
- Add colors, fonts, and animations.
- Maintain consistency across webpages.
- Separate webpage structure from presentation.

Some of the HTML tags are as follows:

Document structure tag:

- **HTML tag:** It is the root of the HTML document which is used to specify that the document is HTML.
Syntax: `<html> Statements... </html>`
- **Head tag:** The head tag is used to contain all the head elements in the HTML file. It contains the title, style, meta, ... etc tag.
Syntax: `<head> Statements... </head>`
- **Body tag:** It is used to define the body of an HTML document. It contains images, tables, lists, ... etc.
Syntax: `<body> Statements... </body>`
- **Title tag:** It is used to define the title of an HTML document. Syntax: `<title> Statements... </title>`

Contentcontainer tag:

- **Heading tag:** It is used to define the heading of an HTML document.

Syntax:

`<h1> Statements... </h>`

`<h2> Statements... </h2>`

`<h3> Statements... </h3>`

<h4> Statements... </h4>

<h5> Statements... </h5>

<h6> Statements... </h6>

- **Paragraph tag:** It is used to define paragraph content in an HTML document. Syntax: <p> Statements... </p>
- **Bold tag:** It is used to specify bold content in an HTML document. Syntax: Statements...
- **Italic tag:** It is used to write the content in italic format. Syntax: <i> Statements... </i>
- **Small (text) tag:** It is used to set the small font size of the content. Syntax: <small> Statements... </small>
- **Underline tag:** It is used to set the content underline. Syntax: <u> Statements... </u>
- **Deleted text tag:** It is used to represent deleted text. It crosses the text content. Syntax: <strike> Statements... </strike>
- **Anchor tag:** It is used to link one page to another page. Syntax: Statements...
- **List tag:** It is used to list the content. Syntax: Statements...
- **Center tag:** It is used to set the content into the center. Syntax: <center>
- **Image Tag**
 - Used to display images.
 - Syntax:
- **Form Tag**
 - Used to create forms.
 - Syntax: <form> Input Fields </form>

Transform Property

The transform property is used to apply visual effects such as rotate, scale, skew, and translate to webpage elements.

Syntax: transform: scale(1.1);

Translate Property

The translate() function is used to move elements horizontally and vertically.

Syntax: transform: translate(20px, 10px);

transform: translate(20px, 10px);

Parameters:

- tx → Horizontal movement
- ty → Vertical movement

Few CSS properties are mentioned below:

- Display Property Defines how elements appear on the

webpage.

- Width and Height Used to set webpage element dimensions.
- Font Family Used to change text fonts.
- Font Size Defines the text size.
- Font Weight Defines text thickness.
- Color Changes text color.
- Background Color Changes background color.
- Padding Creates space inside the element.
- Margin Creates space outside the element.
- Border Adds borders around elements.
- Grid Layout Used for responsive page design.
- Box Shadow Adds shadow effects to webpage elements.

Conclusion:

This experiment demonstrates the development of a complete and visually attractive Pet Shop website using HTML and CSS.

Using HTML, various webpages such as Home, Products, Cart, About Us, Contact, Registration, and Login pages are designed to provide essential functionality for customers and administrators.

Using CSS, the website layout is enhanced with responsive design techniques, CSS Grid layouts, spacing, colors, fonts, and styling effects to improve usability and appearance.

The project helps in understanding the practical implementation of frontend web development technologies and strengthens concepts related to webpage structuring, styling, responsiveness, and user interface design.

This experiment also provides a strong foundation for developing advanced full-stack web applications in future projects.

CODE :

```
:<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Happy Paws Pet Shop</title>
<style>
*{
margin:0;
padding:0;
box-sizing:border-box;
font-family:Arial, sans-serif;}
body{
background:#F5F5DC;
}
/* Header */
header{
background:#6F4E37;
color:white;
```

```

    text-align:center;
    padding:20px;
}
header h1{
    margin-bottom:10px;
}
/* Navigation */
nav{
    background:#3C2F2F;
    padding:15px;
    text-align:center;
}
nav a{
    color:white;
    text-decoration:none;
    margin:0 15px;
    font-size:18px;
    font-weight:bold;
}
nav a:hover{
    color:#F5F5DC;
}
/* Hero */
.hero{
    background:url('https://images.unsplash.com/photo-1450778869180-41d0601e046e');
    background-size:cover;
    background-position:center;
    height:400px;
    display:flex;
    justify-content:center;
    align-items:center;
}
.hero-box{
    background:rgba(0,0,0,0.6);
    color:white;
    padding:30px;
    border-radius:10px;
    text-align:center;
}
.hero-box h1{
    font-size:45px;
}
/* Sections */
section{
    padding:50px 30px;
}
.section-title{
    text-align:center;
    margin-bottom:30px;
}

```

```

    color:#6F4E37;
    font-size:35px;
}
/* Products */
.products{
    display:grid;
    grid-template-columns:repeat(auto-fit,minmax(250px,1fr));
    gap:25px;
}
.card{
    background:white;
    padding:20px;
    border-radius:10px;
    text-align:center;
    box-shadow:0 0 10px gray;
}
.card img{
    width:100%;
    height:220px;
    border-radius:10px;
    object-fit:cover;
}
.card h3{
    margin-top:15px;
}
.price{
    color:#6F4E37;
    font-size:22px;
    font-weight:bold;
    margin:10px 0;
}
button{
    background:#6F4E37;
    color:white;
    border:none;
    padding:10px 20px;
    border-radius:5px;
    cursor:pointer;
}
button:hover{
    background:#3C2F2F;
}
/* Cart */
.cart-box{
    background:white;
    padding:20px;
    border-radius:10px;
    box-shadow:0 0 10px gray;
}

```

```
.cart-item{
  display:flex;
  justify-content:space-between;
  margin:10px 0;
  padding:10px;
  border-bottom:1px solid #ccc;
}
/* About */
.about{
  background:white;
  padding:30px;
  border-radius:10px;
  line-height:1.8;
  box-shadow:0 0 10px gray;
}
/* Forms */
.form-box{
  background:white;
  max-width:400px;
  margin:auto;
  padding:25px;
  border-radius:10px;
  box-shadow:0 0 10px gray;
}
.form-box input{
  width:100%;
  padding:12px;
  margin:10px 0;
  border:1px solid #ccc;
  border-radius:5px;
}
.form-box textarea{
  width:100%;
  padding:12px;
  margin:10px 0;
  border:1px solid #ccc;
  border-radius:5px;
  resize:none;
}
/* Footer */
footer{
  background:#6F4E37;
  color:white;
  text-align:center;
  padding:20px;
}
/* Hide Sections */
.hidden{
  display:none;
}
```



```

}
/* Responsive */
@media(max-width:768px){
  nav a{
    display:block;
    margin:10px 0;
  }
  .hero-box h1{
    font-size:30px;
  }
}
</style>
<script>
var total = 0;
var registeredUsername = "";
var registeredPassword = "";
var isLoggedIn = false;
/* Add To Cart */
function addToCart(product, price){
  if(isLoggedIn == false){
    alert("Please Login First");
    return;
  }
  var cartItems = document.getElementById("cartItems");
  var item = document.createElement("div");
  item.className = "cart-item";
  item.innerHTML =
    "<span>" + product + "</span>" +
    "<span>₹" + price + "</span>";
  cartItems.appendChild(item);
  total = total + price;
  document.getElementById("total").innerHTML = total;
  alert(product + " Added To Cart");
}
/* Registration Validation */
function validateRegister(){
  var name = document.getElementById("name").value;
  var email = document.getElementById("email").value;
  var user = document.getElementById("username").value;
  var pass = document.getElementById("password").value;
  if(name==""){
    alert("Enter Name");
    return false;
  }
  if(email==""){
    alert("Enter Email");
    return false;
  }
  if(email.indexOf("@")==-1){

```

```

    alert("Invalid Email");
    return false;
}
if(user==""){
    alert("Enter Username");
    return false;
}
if(user.length < 4){
    alert("Username must contain minimum 4 characters");
    return false;
}
if(pass==""){
    alert("Enter Password");
    return false;
}
if(pass.length < 6){
    alert("Password must contain minimum 6 characters");
    return false;
}
registeredUsername = user;
registeredPassword = pass;
alert("Registration Successful");
document.getElementById("registerForm").reset();
return false;
}
/* Login Validation */
function validateLogin(){
    var user = document.getElementById("loginuser").value;
    var pass = document.getElementById("loginpass").value;
    if(user==""){
        alert("Enter Username");
        return false;
    }
    if(pass==""){
        alert("Enter Password");
        return false;
    }
    if(user == registeredUsername && pass == registeredPassword){
        isLoggedIn = true;
        alert("Login Successful");
        document.getElementById("products").classList.remove("hidden");
        document.getElementById("cart").classList.remove("hidden");
        document.getElementById("about").classList.remove("hidden");
        document.getElementById("contact").classList.remove("hidden");
    }
    else{
        alert("Invalid Username or Password");
    }
    return false;
}

```

```

}
/* Restrict Access */
function checkLogin(){
    if(isLoggedIn == false){
        alert("Please Login First");
        location.href = "#login";
    }
}
</script>
</head>
<body>
<!-- Header -->
<header>
    <h1> Happy Paws Pet Shop</h1>
    <p>Your One Stop Pet Destination</p>
</header>
<!-- Navigation -->
<nav>
    <a href="#home">Home</a>
    <a href="#products" onclick="checkLogin()">Products</a>
    <a href="#cart" onclick="checkLogin()">Cart</a>
    <a href="#about" onclick="checkLogin()">About</a>
    <a href="#contact" onclick="checkLogin()">Contact</a>
    <a href="#register">Register</a>
    <a href="#login">Login</a>
</nav>
<!-- Hero -->
<section class="hero" id="home">
    <div class="hero-box">
        <h1>Find Your Perfect Pet Companion</h1>
        <p>Pets, Food & Accessories In One Place</p>
    </div>
</section>
<!-- Products -->
<section id="products" class="hidden">
    <h2 class="section-title">Our Products</h2>
    <div class="products">
        <div class="card">
            
            <h3>Dogs</h3>
            <p>Friendly and loyal companions.</p>
            <p class="price">₹15,000</p>
            <button onclick="addToCart('Dog',15000)">
                Add To Cart
            </button>
        </div>
        <div class="card">
            
            <h3>Cats</h3>

```

```

    <p>Playful and adorable pets.</p>
    <p class="price">₹10,000</p>
    <button onclick="addToCart('Cat',10000)">
        Add To Cart
    </button>
</div>
<div class="card">
    
    <h3>Dog Food</h3>
    <p>Healthy premium dog food.</p>
    <p class="price">₹1,200</p>
    <button onclick="addToCart('Dog Food',1200)">
        Add To Cart
    </button>
</div>
</div>
</section>
<!-- Cart -->
<section id="cart" class="hidden">
    <h2 class="section-title">Shopping Cart</h2>
    <div class="cart-box">
        <div id="cartItems">
        </div>
        <h3 style="margin-top:20px;">
            Total Amount : ₹<span id="total">0</span>
        </h3>
    </div>
</section>
<!-- About -->
<section id="about" class="hidden">
    <h2 class="section-title">About Us</h2>
    <div class="about">
        <p>
            Happy Paws Pet Shop provides high-quality pets, food, toys, grooming products, and accessories for all
            pet lovers.
        </p>
        <br>
        <p>
            Our mission is to provide happiness, care, and comfort for pets and their owners.
        </p>
    </div>
</section>
<!-- Contact -->
<section id="contact" class="hidden">
    <h2 class="section-title">Contact Us</h2>
    <div class="form-box">
        <form>
            <input type="text" placeholder="Enter Name">
            <input type="email" placeholder="Enter Email">
            <textarea rows="5" placeholder="Enter Message"></textarea>

```

```

        <button type="submit">Send Message</button>
    </form>
</div>
</section>
<!-- Registration -->
<section id="register">
    <h2 class="section-title">Register</h2>
    <div class="form-box">
        <form id="registerForm" onsubmit="return validateRegister()">
            <input type="text" id="name" placeholder="Full Name">
            <input type="text" id="email" placeholder="Email">
            <input type="text" id="username" placeholder="Username">
            <input type="password" id="password" placeholder="Password">
            <button type="submit">Register</button>
        </form>
    </div>
</section>
<!-- Login -->
<section id="login">
    <h2 class="section-title">Login</h2>
    <div class="form-box">
        <form onsubmit="return validateLogin()">
            <input type="text" id="loginuser" placeholder="Username">
            <input type="password" id="loginpass" placeholder="Password">
            <button type="submit">Login</button>
        </form>
    </div>
</section>
<!-- Footer -->
<footer>
    <p>© 2026 Happy Paws Pet Shop | Contact: 9876543210</p>
</footer>
</body>
</html>

```

Output (Screenshots):

The image displays three sequential screenshots of a web application interface, specifically the 'Sports Store Website 2026'. Each screenshot shows a different page: Login, Register, and Contact. The browser's address bar indicates the URL is 127.0.0.1:3000/index.html?vscod-livepreview=true. The navigation bar at the top of each page includes links for Home, Products, Cart, Contact, Register, and Login. The 'Cart: 5' indicator is visible in the top right corner of each page.

Login Form:

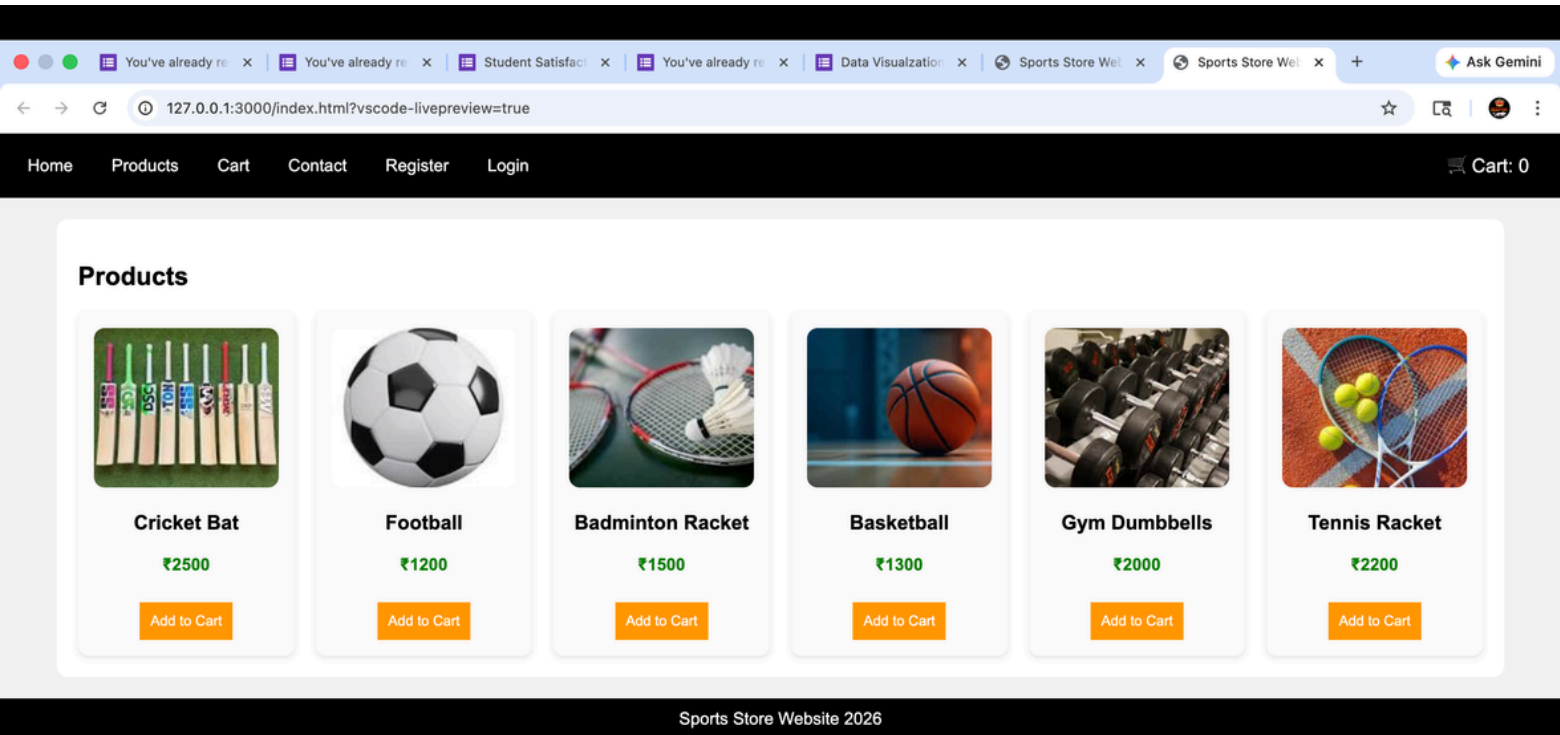
- Fields: Email, Password, User (dropdown menu).
- Button: Login.

Register Form:

- Fields: Name, Email, Password, User (dropdown menu).
- Button: Register.

Contact Form:

- Fields: Name, Email, Message.
- Button: Send.



Topic: Front End Development and Advanced Front End Development

Experiment No.5,6

Problem Statement:

5. JavaScript

Implement user registration and login forms for the Pet Shop website. These forms will allow customers to create an account, log in, and access features such as adding products to the cart and viewing pet product details.

The User Registration Form will allow new users to sign up by entering details such as:

- Full Name
- Email Address
- Username
- Password

The User Login Form will allow registered users to log into the Pet Shop website using:

- Username/Email
- Password

B.

Provide validations for user registration and login forms to ensure:

- All mandatory fields are filled.
- Email format is valid.
- Username contains minimum required characters.
- Password length is secure and valid.

(Contents beyond syllabus)

C.

Develop cart functionality for the Pet Shop website to allow users to:

- Add pet products to the cart.
- Update quantities.
- Remove items from the cart.
- Display total cart amount dynamically.

6. JavaScript

A.

The User Login Form should authenticate registered users using username/email and password.

B.

If login is successful, the user should be redirected to the Pet Shop homepage or dashboard.
(Contents beyond syllabus)

C.

Use localStorage or sessionStorage to store authentication details such as:

- Username
- Login status

This helps users remain logged in even after refreshing or reopening the webpage.
(Contents beyond syllabus)

D.

Store cart data using localStorage whenever:

- Products are added
- Quantities are updated
- Products are removed

Retrieve the stored cart data automatically when the webpage reloads

Objective:

To learn the concepts and implementation of JavaScript for developing interactive and dynamic web applications in the Pet Shop website.

Theory:**What is JavaScript?**

JavaScript is a lightweight, interpreted scripting language used to create dynamic and interactive webpages. It is mainly used for:

- Form validations
- Dynamic content updates
- Cart functionality
- User authentication
- Interactive webpage behavior

JavaScript enhances user experience by making webpages responsive and interactive without reloading the page.

Why JavaScript is Called a Client-Side Scripting Language?

JavaScript is known as a client-side scripting language because it executes directly in the user's browser instead of the server.

When a user opens a website:

- HTML provides webpage structure.
- CSS provides webpage styling.
- JavaScript provides webpage interactivity.

The browser downloads and executes JavaScript on the client side.

Unlike PHP, which executes on the server, JavaScript works on the user's system and responds instantly to user actions

What is a Scripting Language?

A scripting language is a programming language that executes instructions line by line without compilation.

Scripting languages are mainly used to:

- Automate tasks
- Create interactive websites
- Handle user events dynamically

JavaScript is interpreted directly by the browser.

JavaScript Functions Used in the Pet Shop Website

1. getElementById()

The getElementById() method is used to access an HTML element using its unique ID.

Syntax: `document.getElementById("demo");`

Example : `document.getElementById("title").style.color = "red";`

2. innerHTML()

The innerHTML property is used to set or retrieve HTML content inside an element.

Syntax: `element.innerHTML = "Content";`

Example : `document.getElementById("cart").innerHTML = "Item Added";`

3. value

The value property retrieves or sets input field values.

Syntax: `document.getElementById("username").value;`

Example : `var user = document.getElementById("username").value;`

4. parseInt()

The parseInt() function converts string values into integer values.

Syntax: `parseInt(string);`

Example : `var total = parseInt(price1) + parseInt(price2);`

DOM (Document Object Model)

DOM stands for Document Object Model.

It is an interface that represents HTML documents as a tree structure where each HTML element is represented as a node.

Using DOM, JavaScript can:

- Add elements
- Remove elements
- Modify webpage content
- Change styles dynamically

Features of DOM

- Represents webpage structure as nodes.
- Allows manipulation of webpage content dynamically.
- Helps in creating interactive webpages.

Node Types in DOM

1. ELEMENT_NODE
2. ATTRIBUTE_NODE
3. TEXT_NODE
4. COMMENT_NODE
5. DOCUMENT_NODE

Example of DOM Node

```
if(someNode.nodeType == 1)
{
    alert("Node is an element");
}
```

Properties of Node

nodeName

Returns the name of the node.

nodeValue

Returns the value of the node.

childNodes

Returns all child nodes of an element.

Cart Functionality Using JavaScript

The cart system in the Pet Shop website allows users to:

- Add products to cart
- View total amount
- Remove products
- Store cart data in localStorage

This improves user experience by maintaining shopping data even after page refresh.

localStorage in JavaScript

localStorage is used to store webpage data permanently in the browser.

Example: localStorage.setItem("username","Gaurav");

Retrieve Data: localStorage.getItem("username");

sessionStorage in JavaScript

sessionStorage stores data temporarily until the browser tab is closed.

Example: sessionStorage.setItem("login","true");

Conclusion:

This experiment demonstrates the implementation of JavaScript to create a dynamic and interactive Pet Shop website.

By implementing user registration and login systems, the website provides secure authentication and personalized access for customers. JavaScript validations improve data accuracy and ensure proper user input handling.

The cart functionality enables users to add, update, and remove products dynamically, creating an efficient online shopping experience. Using localStorage and sessionStorage allows user authentication and cart details to persist across sessions and page reloads.

Overall, this assignment highlights the importance of JavaScript in developing modern web applications with improved interactivity, responsiveness, and user experience for the Pet Shop website.

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Happy Paws Pet Shop</title>
<style>
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:Arial, sans-serif;
}
body{
    background:#F5F5DC;
}
/* Header */
header{
    background:#6F4E37;
    color:white;
    text-align:center;
    padding:20px;
}
header h1{
    margin-bottom:10px;
}
/* Navigation */
nav{
    background:#3C2F2F;
    padding:15px;
    text-align:center;
}
nav a{
    color:white;
    text-decoration:none;
    margin:0 15px;
    font-size:18px;
    font-weight:bold;
}
nav a:hover{
    color:#F5F5DC;
}
/* Hero */
.hero{
    background:url('https://images.unsplash.com/photo-1450778869180-41d0601e046e');
    background-size:cover;
    background-position:center;
```

```

height:400px;
display:flex;
justify-content:center;
align-items:center;
}
.hero-box{
background:rgba(0,0,0,0.6);
color:white;
padding:30px;
border-radius:10px;
text-align:center;
}
.hero-box h1{
font-size:45px;
}
/* Sections */
section{
padding:50px 30px;
}
.section-title{
text-align:center;
margin-bottom:30px;
color:#6F4E37;
font-size:35px;
}
/* Products */
.products{
display:grid;
grid-template-columns:repeat(auto-fit,minmax(250px,1fr));
gap:25px;
}
.card{
background:white;
padding:20px;
border-radius:10px;
text-align:center;
box-shadow:0 0 10px gray;
}
.card img{
width:100%;
height:220px;
border-radius:10px;
object-fit:cover;
}
.card h3{
margin-top:15px;
}
.price{
color:#6F4E37;

```

```

font-size:22px;
font-weight:bold;
margin:10px 0;
}
button{
background:#6F4E37;
color:white;
border:none;
padding:10px 20px;
border-radius:5px;
cursor:pointer;
}
button:hover{
background:#3C2F2F;
}
/* Cart */
.cart-box{
background:white;
padding:20px;
border-radius:10px;
box-shadow:0 0 10px gray;
}
.cart-item{
display:flex;
justify-content:space-between;
align-items:center;
border-bottom:1px solid #ccc;
padding:10px 0;
}
.remove-btn{
background:red;
padding:5px 10px;
border:none;
color:white;
border-radius:5px;
cursor:pointer;
}
/* About */
.about{
background:white;
padding:30px;
border-radius:10px;
line-height:1.8;
box-shadow:0 0 10px gray;
}
/* Forms */
.form-box{
background:white;
max-width:400px;

```

```

margin:auto;
padding:25px;
border-radius:10px;
box-shadow:0 0 10px gray;
}
.form-box input,
.form-box textarea{
width:100%;
padding:12px;
margin:10px 0;
border:1px solid #ccc;
border-radius:5px;

.form-box textarea{
resize:none;
}
/* Footer */
footer{
background:#6F4E37;
color:white;
text-align:center;
padding:20px;
}
/* Hidden */
.hidden{
display:none;
}
@media(max-width:768px){
nav a{
display:block;
margin:10px 0;
}
.hero-box h1{
font-size:30px;
}
}
</style>
<script>
/* Local Storage Check */
window.onload = function(){
if(localStorage.getItem("loginStatus") == "true"){
document.getElementById("products").classList.remove("hidden");
document.getElementById("cart").classList.remove("hidden");
document.getElementById("about").classList.remove("hidden");
document.getElementById("contact").classList.remove("hidden");
loadCart();
}
}
/* Registration */

```

```
function validateRegister(){
    var name = document.getElementById("name").value;
    var email = document.getElementById("email").value;
    var username = document.getElementById("username").value;
    var password = document.getElementById("password").value;
    if(name==""){
        alert("Enter Name");
        return false;
    }
    if(email==""){
        alert("Enter Email");
        return false;
    }
    if(email.indexOf("@")==-1){
        alert("Invalid Email");
        return false;
    }
    if(username==""){
        alert("Enter Username");
        return false;
    }
    if(username.length < 4){
        alert("Username must contain minimum 4 characters");
        return false;
    }
    if(password==""){
        alert("Enter Password");
        return false;
    }
    if(password.length < 6){
        alert("Password must contain minimum 6 characters");
        return false;
    }
    localStorage.setItem("username", username);
    localStorage.setItem("password", password);
    alert("Registration Successful");
    document.getElementById("registerForm").reset();
    return false;
}

/* Login */
function validateLogin(){
    var loginuser = document.getElementById("loginuser").value;
    var loginpass = document.getElementById("loginpass").value;
    var savedUser = localStorage.getItem("username");
    var savedPass = localStorage.getItem("password");
    if(loginuser==""){
        alert("Enter Username");
        return false;
    }
}
```



```

if(loginpass==""){
    alert("Enter Password");
    return false;
}
if(loginuser == savedUser && loginpass == savedPass){
    localStorage.setItem("loginStatus","true");
    alert("Login Successful");
    document.getElementById("products").classList.remove("hidden");
    document.getElementById("cart").classList.remove("hidden");
    document.getElementById("about").classList.remove("hidden");
    document.getElementById("contact").classList.remove("hidden");
}
else{
    alert("Invalid Username or Password");
}
return false;
}
/* Logout */
function logout(){
    localStorage.removeItem("loginStatus");
    alert("Logged Out");
    location.reload();
}
/* Restrict Access */
function checkLogin(){
    if(localStorage.getItem("loginStatus") != "true"){
        alert("Please Login First");
        location.href="#login";
        return false;
    }
}
/* Cart Array */
var cart = [];
/* Add To Cart */
function addToCart(product, price){
    if(localStorage.getItem("loginStatus") != "true"){
        alert("Please Login First");
        return;
    }
    var item = {
        product:product,
        price:price
    };
    cart.push(item);
    localStorage.setItem("cartData", JSON.stringify(cart));
    displayCart();
    alert(product + " Added To Cart");
}
/* Display Cart */

```

```
function displayCart(){
    var cartItems = document.getElementById("cartItems");
    cartItems.innerHTML = "";
    var total = 0;
    for(var i=0; i<cart.length; i++){
        total = total + cart[i].price;
        cartItems.innerHTML +
        "<div class='cart-item'>" +
        "<span>" + cart[i].product + " - ₹" + cart[i].price + "</span>" +
        "<button class='remove-btn' onclick='removeItem(" + i + ")'>Remove</button>" +
        "</div>";
    }
    document.getElementById("total").innerHTML = total;
}
/* Load Cart */
function loadCart(){
    var savedCart = localStorage.getItem("cartData");
    if(savedCart != null){
        cart = JSON.parse(savedCart);
        displayCart();
    }
}
/* Remove Item */
function removeItem(index){
    cart.splice(index,1);
    localStorage.setItem("cartData", JSON.stringify(cart));
    displayCart();
}
</script>
</head>
<body>
<header>
    <h1> Happy Paws Pet Shop</h1>
    <p>Your One Stop Pet Destination</p>
</header>
<nav>
    <a href="#home">Home</a>
    <a href="#products" onclick="checkLogin()">Products</a>
    <a href="#cart" onclick="checkLogin()">Cart</a>
    <a href="#about" onclick="checkLogin()">About</a>
    <a href="#contact" onclick="checkLogin()">Contact</a>
    <a href="#register">Register</a>
    <a href="#login">Login</a>
    <a href="#" onclick="logout()">Logout</a>
</nav>
<!-- Hero -->
<section class="hero" id="home">
    <div class="hero-box">
        <h1>Find Your Perfect Pet Companion</h1>
```

```

    <p>Pets, Food & Accessories In One Place</p>
</div>
</section>
<!-- Products -->
<section id="products" class="hidden">
    <h2 class="section-title">Our Products</h2>
    <div class="products">
        <div class="card">
            
            <h3>Dogs</h3>
            <p>Friendly and loyal companions.</p>
            <p class="price">₹15,000</p>
            <button onclick="addToCart('Dog',15000)">
                Add To Cart
            </button>
        </div>
        <div class="card">
            
            <h3>Cats</h3>
            <p>Playful and adorable pets.</p>
            <p class="price">₹10,000</p>
            <button onclick="addToCart('Cat',10000)">
                Add To Cart
            </button>
        </div>
        <div class="card">
            
            <h3>Dog Food</h3>
            <p>Healthy premium dog food.</p>
            <p class="price">₹1,200</p>
            <button onclick="addToCart('Dog Food',1200)">
                Add To Cart
            </button>
        </div>
    </div>
</section>
<!-- Cart -->
<section id="cart" class="hidden">
    <h2 class="section-title">Shopping Cart</h2>
    <div class="cart-box">
        <div id="cartItems">
            </div>
            <h3 style="margin-top:20px;">
                Total Amount : ₹<span id="total">0</span>
            </h3>
        </div>
    </section>
<!-- About -->
<section id="about" class="hidden">

```

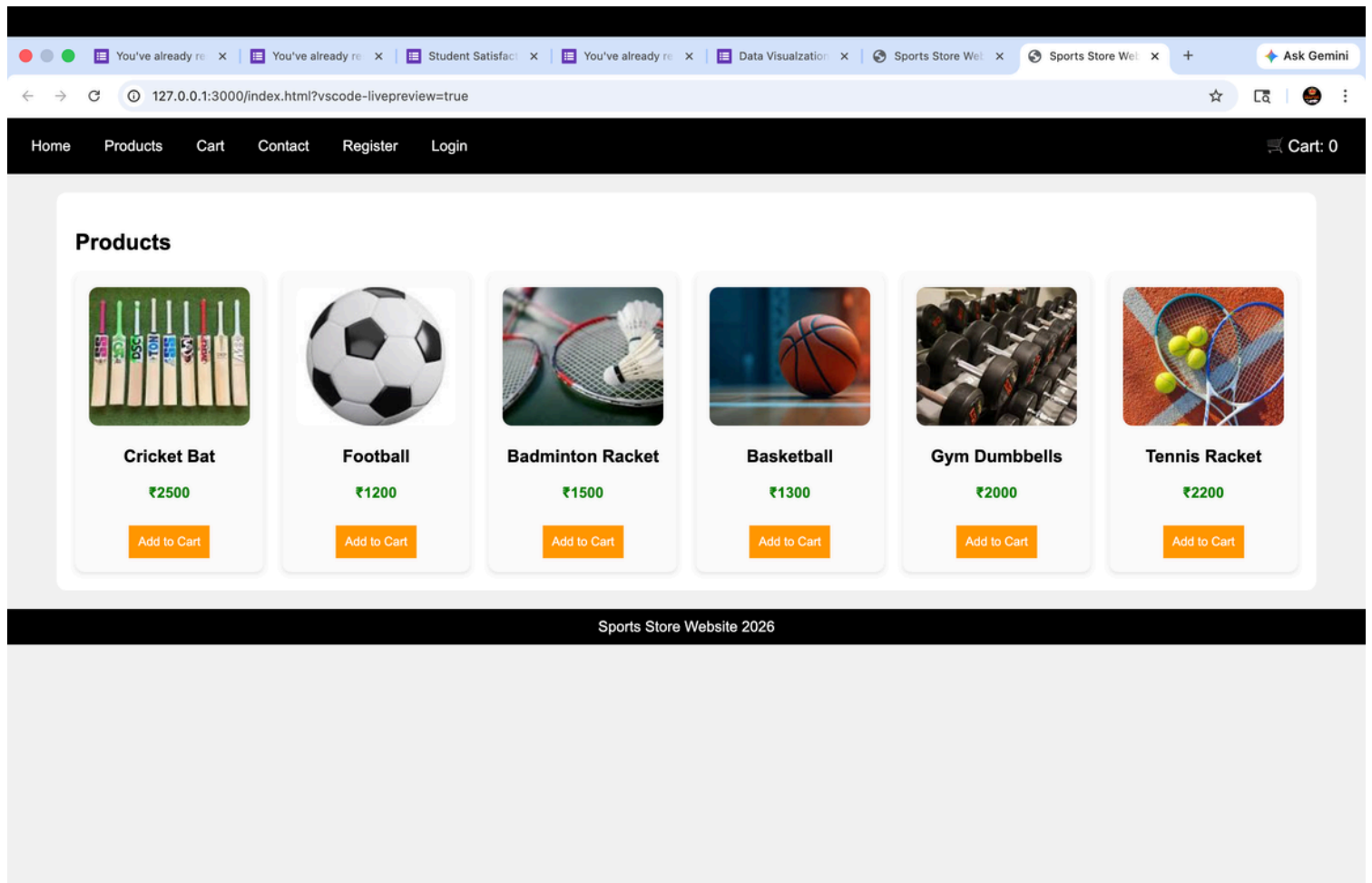
```

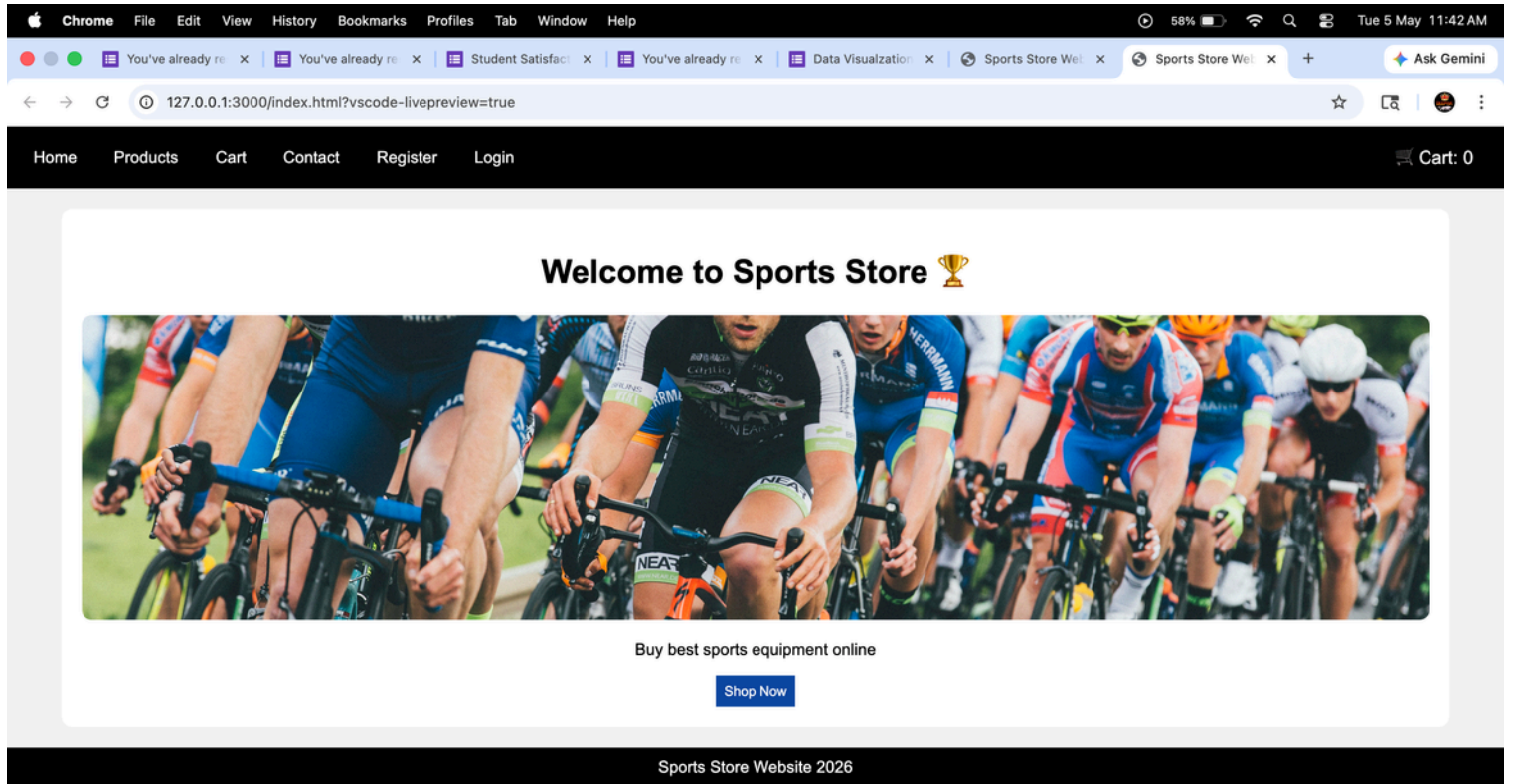
<h2 class="section-title">About Us</h2>
<div class="about">
  <p>
    Happy Paws Pet Shop provides high-quality pets, food, toys, grooming products, and accessories for all
    pet lovers.
  </p>
  <br>
  <p>
    Our mission is to provide happiness, care, and comfort for pets and their owners.
  </p>
</div>
</section>
<!-- Contact -->
<section id="contact" class="hidden">
  <h2 class="section-title">Contact Us</h2>
  <div class="form-box">
    <form>
      <input type="text" placeholder="Enter Name">
      <input type="email" placeholder="Enter Email">
      <textarea rows="5" placeholder="Enter Message"></textarea>
      <button type="submit">Send Message</button>
    </form>
  </div>
</section>
<!-- Register -->
<section id="register">
  <h2 class="section-title">Register</h2>
  <div class="form-box">
    <form id="registerForm" onsubmit="return validateRegister()">
      <input type="text" id="name" placeholder="Full Name">
      <input type="text" id="email" placeholder="Email">
      <input type="text" id="username" placeholder="Username">
      <input type="password" id="password" placeholder="Password">
      <button type="submit">Register</button>
    </form>
  </div>
</section>
<!-- Login -->
<section id="login">
  <h2 class="section-title">Login</h2>
  <div class="form-box">
    <form onsubmit="return validateLogin()">
      <input type="text" id="loginuser" placeholder="Username">
      <input type="password" id="loginpass" placeholder="Password">
      <button type="submit">Login</button>
    </form>
  </div>
</section>
<footer>
  <p>© 2026 Happy Paws Pet Shop | Contact: 9876543210</p>

```

```
</footer>
</body>
</html>
```

Output (Screenshots):





Topic: Server Side Scripting and Working with database and Web Application Development

Experiment No:7,8,9,10

7. PHP

A. Develop a PHP script to handle user registration for the Pet Shop website. The script should accept user details such as:

- Full Name
- Email Address
- Username
- Password

The registration system should store user information securely for future login authentication.

B. Implement error handling and validations during registration to notify users about:

- Empty fields
- Invalid email format
- Weak passwords
- Existing usernames

C. Provide feedback to users after successful registration through:

- Success message
- Redirect to Login Page
- Confirmation alert

8. PHP

A. Develop a PHP script to handle user login for the Pet Shop website. The script should authenticate users using:

- Username/Email
- Password

B. Provide feedback to users upon successful login through:

- Welcome message
- Redirect to Homepage
- User Dashboard access

C. Implement error handling to notify users about:

- Incorrect username/password
- Empty login fields
- Unauthorized access

D. Display a personalized welcome message after successful login.

Example: Welcome, Sujal!

9. PHP and MySQL

A. Develop a PHP script for managing the shopping cart of the Pet Shop website.

The shopping cart should allow users to:

- Add pet products to cart
- View cart items
- Remove items from cart
- Update product quantities

B. Develop a PHP and MySQL based cart management system where:

- Cart data is stored in MySQL database
- Cart remains persistent across sessions
- Product information is retrieved dynamically from database

The system should support:

- Insert cart items
- Delete cart items
- Display cart contents
- Calculate total amount

10.PHP and MySQL

A.Develop a PHP script to handle the checkout process for users ready to purchase products from the PetShop website.

The checkout system should:

- Process cart data
- Validate customer information
- Display order confirmation
- Handle checkout errors

B . Develop a PHP and MySQL integrated checkout system that:

- Stores order information in database
- Retrieves customer details
- Processes shopping cart records
- Confirms successful order placement

The system should provide:

- Success messages
- Order summaries
- Error handling for failed transactions

Objective:

PHP:

To understand the implementation of PHP for server-side scripting in the Pet Shop website, enabling dynamic webpage generation, user authentication, cart management, and checkout processing.

MySQL:

To understand database management using MySQL for storing and retrieving user information, cart details, product records, and customer orders efficiently.

Theory:

What is PHP?

PHP (Hypertext Preprocessor) is an open-source server-side scripting language used to create dynamic and interactive web applications.

PHP scripts execute on the server and send the generated output to the browser in the form of HTML pages.

PHP is widely used for:

- User authentication
- Form handling
- Database interaction
- Session management
- Dynamic webpage generation

PHP was originally developed by Rasmus Lerdorf in 1994.

Features of PHP

- Open-source
- Platform independent
- Supports database connectivity
- Easy integration with HTML
- Fast and efficient
- Supports session handling

Data Types in PHP

PHP supports different types of data:

1. Integer
2. Float
3. String
4. Boolean
5. Array
6. Object
7. NULL
8. Resource

PHP Arrays

Arrays are used to store multiple values under a single variable name.

Types of Arrays in PHP

1. Indexed Array

Stores values using numeric indexes.

Example: `$products = array("Dog Food", "Cat Toy", "Pet Shampoo");`

2. Associative Array

Stores values using named keys.

Example: `$user = array("name"=>"Gaurav", "email"=>"gaurav@gmail.com");`

3. Multidimensional Array

Contains arrays inside another array.

```
Example: $cart = array(  
array("Dog Food",1200),  
array("Cat Toy",500)  
);
```

MySQL Database

MySQL is a relational database management system used to:

- Store user information
- Store cart records
- Store product details
- Manage orders

It helps websites maintain data permanently.

PHP Database Connection

To connect PHP with MySQL database:

- Start Apache and MySQL in XAMPP
- Open phpMyAdmin
- Create database
- Connect using PHP script

```
Example: $conn = mysqli_connect("localhost","root","","petshop");
```

XAMPP Server

XAMPP is a local web server environment used to run PHP and MySQL applications locally.

It includes:

- Apache Server
- MySQL Database
- PHP Interpreter
- phpMyAdmin

Session Handling in PHP

Sessions are used to store user data temporarily across webpages.

```
Example: session_start();  
$_SESSION['username'] = "Sujal";
```

Shopping Cart in PHP

The cart functionality in the Pet Shop website allows users to:

- Add products
- Remove products
- Update quantities
- Store cart data in database

This creates a real-time e-commerce experience.

Checkout Process

The checkout module validates:

- User details

- Cart items
- Payment information

After successful checkout:

- Order details are stored in MySQL
- Confirmation message is displayed

Error Handling in PHP

PHP handles errors using conditions and validation checks.

```
Example: if(empty($username)){
echo "Username Required";
}
```

Conclusion:

This experiment demonstrates the implementation of PHP and MySQL for developing a dynamic and database-driven Pet Shop website.

Using PHP, functionalities such as:

- User Registration
- User Login
- Shopping Cart Management
- Checkout Processing
- Session Handling

are implemented successfully.

Using MySQL, the website stores:

- User records
- Product details
- Cart information
- Customer orders

This experiment helps in understanding server-side scripting, database management, validation handling, and e-commerce website development concepts practically.

The integration of PHP and MySQL creates a complete dynamic Pet Shop web application with secure authentication, persistent cart management, and order processing functionality.

CODE :

```
<?php
session_start();
/* Database Simulation Using Arrays */
$users = array();
$products = array(
    array("Dog","15000","https://images.unsplash.com/photo-1517849845537-4d257902454a"),
    array("Cat","10000","https://images.unsplash.com/photo-1511044568932-338cba0ad803"),
    array("Dog Food","1200","https://images.unsplash.com/photo-1583511655857-d19b40a7a54e"));
/* Registration */
$registerMessage = "";
if(isset($_POST['register'])){
    $name = $_POST['name'];
    $email = $_POST['email'];
```

```

$username = $_POST['username'];
$password = $_POST['password'];
if($name == "" || $email == "" || $username == "" || $password == ""){
    $registerMessage = "All Fields Required";}
elseif(!filter_var($email, FILTER_VALIDATE_EMAIL)){
    $registerMessage = "Invalid Email Format";}
elseif(strlen($password) < 6){
    $registerMessage = "Password must contain minimum 6 characters";}
else{
    $_SESSION['username'] = $username;
    $_SESSION['password'] = $password;
    $registerMessage = "Registration Successful";}}

/* Login */
$loginMessage = "";
if(isset($_POST['login'])){
    $loginuser = $_POST['loginuser'];
    $loginpass = $_POST['loginpass'];
    if($loginuser == $_SESSION['username'] && $loginpass == $_SESSION['password']){
        $_SESSION['loginStatus'] = true;
        $loginMessage = "Welcome " . $loginuser;}
    else{
        $loginMessage = "Invalid Username or Password";}}

/* Logout */
if(isset($_GET['logout'])){
    session_destroy();
    header("Location: ".$_SERVER['PHP_SELF']);}

/* Cart */
if(!isset($_SESSION['cart'])){
    $_SESSION['cart'] = array();}

/* Add To Cart */
if(isset($_GET['add'])){
    $product = $_GET['product'];
    $price = $_GET['price'];
    $item = array(
        "product"=>$product,
        "price"=>$price);
    array_push($_SESSION['cart'],$item);}

/* Remove Cart Item */
if(isset($_GET['remove'])){
    $index = $_GET['remove'];
    unset($_SESSION['cart'][$index]);
    $_SESSION['cart'] = array_values($_SESSION['cart']);}

/* Checkout */
$checkoutMessage = "";
if(isset($_POST['checkout'])){
    if(count($_SESSION['cart']) == 0){
        $checkoutMessage = "Cart is Empty";}
    else{
        $_SESSION['cart'] = array();
    }
}

```

```

    $checkoutMessage = "Order Placed Successfully";}}?>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Happy Paws Pet Shop</title>
<style>
*{
    margin:0;
    padding:0;
    box-sizing:border-box;
    font-family:Arial;}
body{
    background:#F5F5DC;}
/* Header */
header{
    background:#6F4E37;
    color:white;
    text-align:center;
    padding:20px;}
/* Navigation */
nav{
    background:#3C2F2F;
    padding:15px;
    text-align:center;}
nav a{
    color:white;
    text-decoration:none;
    margin:15px;
    font-weight:bold;}
/* Hero */
.hero{
    background:url('https://images.unsplash.com/photo-1450778869180-41d0601e046e');
    background-size:cover;
    background-position:center;
    height:350px;
    display:flex;
    justify-content:center;
    align-items:center;}
.hero-box{
    background:rgba(0,0,0,0.6);
    color:white;
    padding:30px;
    border-radius:10px;}
/* Sections */
section{
    padding:40px;}
.section-title{

```

```

    text-align:center;
    color:#6F4E37;
    margin-bottom:30px;
    font-size:35px;}

/* Products */
.products{
    display:grid;
    grid-template-columns:repeat(auto-fit,minmax(250px,1fr));
    gap:20px;}
.card{
    background:white;
    padding:20px;
    border-radius:10px;
    text-align:center;
    box-shadow:0 0 10px gray;}
.card img{
    width:100%;
    height:220px;
    object-fit:cover;
    border-radius:10px;}
.price{
    color:#6F4E37;
    font-size:22px;
    margin:10px 0;}

/* Forms */
.form-box{
    background:white;
    max-width:400px;
    margin:auto;
    padding:25px;
    border-radius:10px;
    box-shadow:0 0 10px gray;}
.form-box input{
    width:100%;
    padding:12px;
    margin:10px 0;
    border:1px solid #ccc;
    border-radius:5px;}

/* Buttons */
button{
    background:#6F4E37;
    color:white;
    border:none;
    padding:10px 20px;
    border-radius:5px;
    cursor:pointer;}

/* Cart */
.cart-box{
    background:white;

```

```
padding:20px;
border-radius:10px;
box-shadow:0 0 10px gray;
}
.cart-item{
display:flex;
justify-content:space-between;
border-bottom:1px solid #ccc;
padding:10px 0;
}
/* Message */
.message{
text-align:center;
color:red;
margin-top:10px;
}
/* Footer */
footer{
background:#6F4E37;
color:white;
text-align:center;
padding:20px;
margin-top:30px;
}
</style>
</head>
<body>
<header>
<h1> Happy Paws Pet Shop</h1>
<p>Your One Stop Pet Destination</p>
</header>
<nav>
<a href="#home">Home</a>
<a href="#products">Products</a>
<a href="#cart">Cart</a>
<a href="#register">Register</a>
<a href="#login">Login</a>
<a href="?logout=true">Logout</a>
</nav>
<!-- Hero -->
<section class="hero" id="home">
<div class="hero-box">
<h1>Find Your Perfect Pet Companion</h1>
<p>Pets, Food & Accessories In One Place</p>
</div>
</section>
<!-- Products -->
<section id="products">
<h2 class="section-title">Our Products</h2>
```



```

<div class="products">
<?php
for($i=0; $i<count($products); $i++){
?>
<div class="card">

<h3><?php echo $products[$i][0]; ?></h3>
<p class="price">₹<?php echo $products[$i][1]; ?></p>
<a href="?add=true&product=<?php echo $products[$i][0]; ?>&price=<?php echo $products[$i][1]; ?>"
<button>Add To Cart</button>
</a>
</div>
<?php}
?>
</div>
</section>
<!-- Cart -->
<section id="cart">
<h2 class="section-title">Shopping Cart</h2>
<div class="cart-box">
<?php
$total = 0;
for($i=0; $i<count($_SESSION['cart']); $i++){
$total = $total + $_SESSION['cart'][$i]['price'];
?>
<div class="cart-item">
<span>
<?php echo $_SESSION['cart'][$i]['product']; ?>
- ₹<?php echo $_SESSION['cart'][$i]['price']; ?>
</span>
<a href="?remove=<?php echo $i; ?>">
<button>Remove</button>
</a>
</div>
<?php}
?>
<h3 style="margin-top:20px;">
Total Amount : ₹<?php echo $total; ?>
</h3>
<br>
<form method="POST">
<button type="submit" name="checkout">
Checkout
</button>
</form>
<p class="message">
<?php echo $checkoutMessage; ?>
</p>
</div>

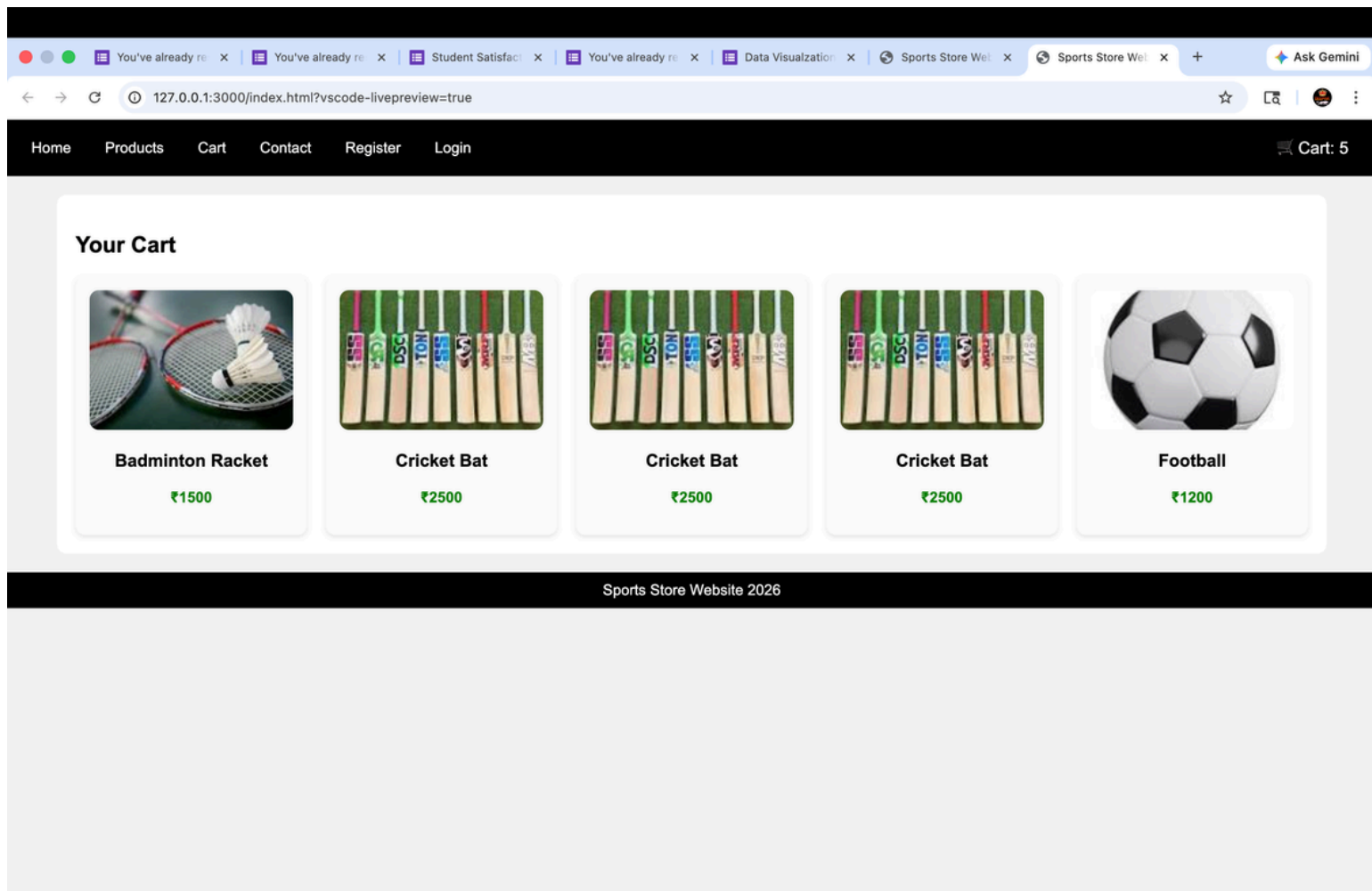
```

```

</section>
<!-- Register -->
<section id="register">
<h2 class="section-title">Register</h2>
<div class="form-box">
<form method="POST">
<input type="text" name="name" placeholder="Full Name">
<input type="text" name="email" placeholder="Email">
<input type="text" name="username" placeholder="Username">
<input type="password" name="password" placeholder="Password">
<button type="submit" name="register">
Register
</button>
</form>
<p class="message">
<?php echo $registerMessage; ?>
</p>
</div>
</section>
<!-- Login -->
<section id="login">
<h2 class="section-title">Login</h2>
<div class="form-box">
<form method="POST">
<input type="text" name="loginuser" placeholder="Username">
<input type="password" name="loginpass" placeholder="Password">
<button type="submit" name="login">
Login
</button>
</form>
<p class="message">
<?php echo $loginMessage; ?>
</p>
</div>
</section>
<footer>
<p>© 2026 Happy Paws Pet Shop | Contact: 9876543210</p>
</footer>
</body>
</html>

```

Output (Screenshots):



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